

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL1- CASE STUDY

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Code of Conduct:

I _____ Venu G _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: SoilDry $\rightarrow$ IrrigationNeeded
- 2) P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: SoilDry = True, CropWheat = True

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell}[1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell}[1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at}[x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understan ding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well- structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Case Study 1 — Smart Medical Diagnosis System

(a) Propositional representation of rules and facts

Symbol definitions:

- F = Patient has Fever
- C = Patient has Cough
- R_s = Patient has Rash
- B = Patient has Breathlessness
- Flu = Patient has Possible Flu
- Mea = Patient has Possible Measles
- $Pneu$ = Patient has Possible Pneumonia

Rules (Horn clauses):

- $R1: F \wedge C \Rightarrow Flu$
- $R2: F \wedge R_s \Rightarrow Mea$
- $R3: C \wedge B \Rightarrow Pneu$

Facts about Patient A:

- $F = \text{True}$, $C = \text{True}$, $B = \text{True}$ (R_s is not asserted, so $R_s = \text{unknown/false}$ by closed-world assumption)

(b) Forward chaining — deriving all possible diagnoses

Initial fact base: $\{ F, C, B \}$

Iteration	Rule checked	Premises satisfied?	Fact derived
1	$R1: F \wedge C \Rightarrow Flu$	$F=\text{True}, C=\text{True} \rightarrow \text{satisfied}$	Flu (added)
1	$R2: F \wedge R_s \Rightarrow Mea$	$F=\text{True}, R_s=\text{unknown} \rightarrow$ NOT satisfied	— (no fact derived)
1	$R3: C \wedge B \Rightarrow Pneu$	$C=\text{True}, B=\text{True} \rightarrow \text{satisfied}$	$Pneu$ (added)
2	$R1, R2, R3$ re-checked	No new premise combinations become true	No new facts — fixed point reached

Final fact base after forward chaining: $\{ F, C, B, Flu, Pneu \}$.

Derived diagnoses for Patient A: Possible Flu and Possible Pneumonia. Possible Measles is NOT derived because the Rash symptom (R_s) was never observed/asserted true, so rule $R2$'s premise is not satisfied.

(c) Backward chaining — verifying "Patient A has Pneumonia"

Goal: Pneu (Pneumonia). Backward chaining looks for a rule whose conclusion is Pneu, then recursively tries to prove each premise of that rule as a sub-goal, until every branch bottoms out in a known fact (AND-OR goal-reduction search).

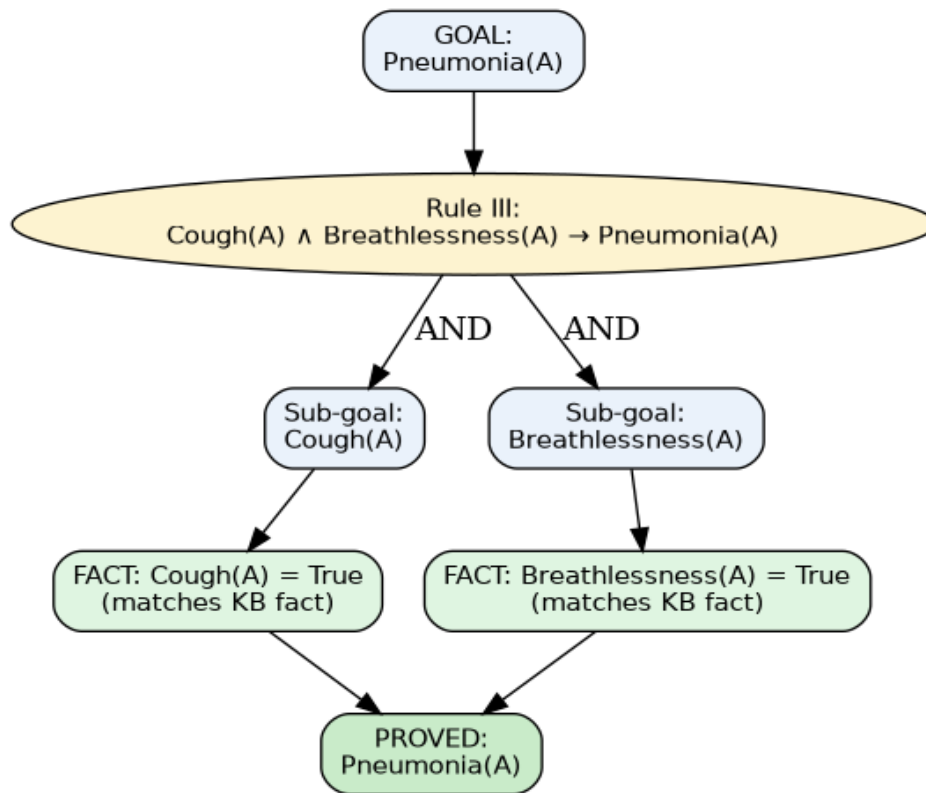


Fig. 1: Goal-reduction (AND) tree for proving *Pneumonia(A)*

Reduction trace:

- Goal: Pneu — matches the conclusion of R3 ($C \wedge B \Rightarrow \text{Pneu}$), so Pneu reduces to the AND of two sub-goals: C and B.
 - Sub-goal C (Cough): found directly as a known fact ($C = \text{True}$). Sub-goal proved.
 - Sub-goal B (Breathlessness): found directly as a known fact ($B = \text{True}$). Sub-goal proved.
- Both AND-branches are proved, so the parent goal Pneu is PROVED.

Conclusion: the goal 'Patient A has Pneumonia' is confirmed by backward chaining, consistent with the forward-chaining result in (b).

Case Study 2 — Autonomous Traffic Management Agent

Rules:

- Rule 1: $\forall x \text{ Vehicle}(x) \wedge \text{EmergencyType}(x) \Rightarrow \text{ClearPath}(x)$
- Rule 2: $\forall x \text{ ClearPath}(x) \Rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$
- Rule 3: $\forall x \forall y \text{ Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \Rightarrow \text{Proceed}(x)$

Facts: $\text{Vehicle}(\text{Ambulance})$, $\text{EmergencyType}(\text{Ambulance})$, $\text{Vehicle}(\text{CarA})$, $\text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Unification of Rule 1 with the facts

Rule 1's premises are $\text{Vehicle}(x)$ and $\text{EmergencyType}(x)$ (the same variable x links both literals, since the rule requires one vehicle that is simultaneously an emergency-type vehicle). We unify each premise literal against the matching fact:

Step 1 — $\text{UNIFY}(\text{Vehicle}(x), \text{Vehicle}(\text{Ambulance}))$

- x is a variable, Ambulance is a constant $\rightarrow$ bind $x/\text{Ambulance}$.
- $\theta_1 = \{ x / \text{Ambulance} \}$

Step 2 — $\text{UNIFY}(\text{EmergencyType}(x), \text{EmergencyType}(\text{Ambulance}))$ under θ_1

- x is already bound to Ambulance under θ_1 ; substituting gives $\text{EmergencyType}(\text{Ambulance})$, which matches the fact exactly.
- $\theta_2 = \theta_1 = \{ x / \text{Ambulance} \}$ (no new binding needed — consistent)

Result: MGU $\theta = \{ x / \text{Ambulance} \}$

Applying θ to Rule 1's conclusion $\text{ClearPath}(x)$ gives $\text{ClearPath}(\text{Ambulance})$ — so the fact $\text{ClearPath}(\text{Ambulance})$ is newly derivable.

(b) Resolution proof that $\text{Proceed}(\text{CarA})$ is true

Step 1 — Clausal (CNF) form of every rule and fact:

Label	Clause (CNF)	Source
C1	$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$	Rule 1
C2a	$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$	Rule 2 (first conjunct)
C2b	$\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$	Rule 2 (second conjunct)
C3	$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$	Rule 3
F1	$\text{Vehicle}(\text{Ambulance})$	Fact
F2	$\text{EmergencyType}(\text{Ambulance})$	Fact

F3	Vehicle(CarA)	Fact
F4	Behind(CarA, Ambulance)	Fact
$\neg G$	$\neg \text{Proceed}(\text{CarA})$	Negated goal

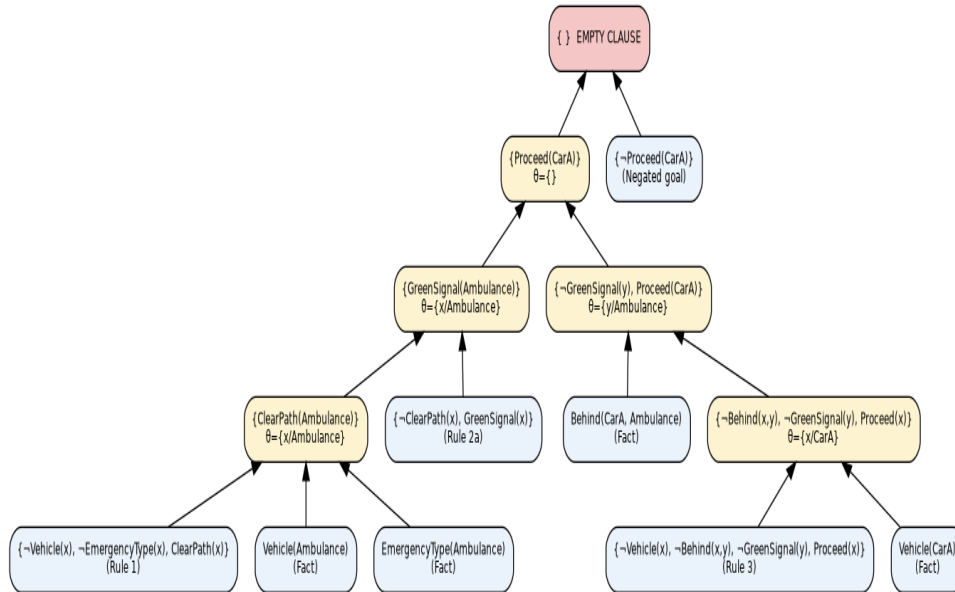


Fig. 2: Resolution refutation tree proving $\text{Proceed}(\text{CarA})$

Step-by-step resolution trace:

- $R1 = \text{resolve}(C1, F1) \text{ with } \theta = \{x/\text{Ambulance}\} \rightarrow \{\neg \text{EmergencyType}(\text{Ambulance}), \text{ClearPath}(\text{Ambulance})\}$
- $R1' = \text{resolve}(R1, F2) \rightarrow \{\text{ClearPath}(\text{Ambulance})\}$
- $R2 = \text{resolve}(C2a, R1') \text{ with } \theta = \{x/\text{Ambulance}\} \rightarrow \{\text{GreenSignal}(\text{Ambulance})\}$
- $R3 = \text{resolve}(C3, F3) \text{ with } \theta = \{x/\text{CarA}\} \rightarrow \{\neg \text{Behind}(\text{CarA}, y), \neg \text{GreenSignal}(y), \text{Proceed}(\text{CarA})\}$
- $R4 = \text{resolve}(R3, F4) \text{ with } \theta = \{y/\text{Ambulance}\} \rightarrow \{\neg \text{GreenSignal}(\text{Ambulance}), \text{Proceed}(\text{CarA})\}$
- $R5 = \text{resolve}(R4, R2) \rightarrow \{\text{Proceed}(\text{CarA})\}$
- $R6 = \text{resolve}(R5, \neg G) \rightarrow \{\} \text{ EMPTY CLAUSE}$

Since the empty clause is derived, $\text{KB} \wedge \neg \text{Proceed}(\text{CarA})$ is unsatisfiable, hence $\text{KB} \models \text{Proceed}(\text{CarA})$ — the theorem is proved: CarA is permitted to proceed.

(c) Is the KB sufficient for two simultaneous emergency vehicles?

No — the current KB is insufficient once more than one emergency vehicle can be present at the same junction at the same time. The reasons and the fixes needed are:

- Rule 2 unconditionally grants $\text{GreenSignal}(x)$ to every x for which $\text{ClearPath}(x)$ holds. If two ambulances from conflicting directions both satisfy Rule 1, Rule 2 would derive GreenSignal for both simultaneously, producing a logically consistent but physically unsafe outcome (two green signals on conflicting approaches).

- There is no predicate capturing priority or conflict between two emergency vehicles (e.g. which one arrived first, which has higher priority class, or which directions physically conflict at the intersection).
- There is no rule preventing $\text{GreenSignal}(x)$ and $\text{GreenSignal}(z)$ from holding together when x and z are on conflicting approaches — the KB has no notion of $\text{Conflicts}(x,z)$ or $\text{MutuallyExclusive}(x,z)$.

Additional predicates/rules needed:

- $\text{Conflicts}(x,y)$ — true if the paths of vehicles x and y intersect at the junction.
- $\text{Priority}(x,n)$ — a priority level n for vehicle x (e.g. based on emergency severity or arrival time).
- New rule: $\forall x \forall y \text{ ClearPath}(x) \wedge \text{ClearPath}(y) \wedge \text{Conflicts}(x,y) \wedge \text{Priority}(x,n_1) \wedge \text{Priority}(y,n_2) \wedge n_1 > n_2 \Rightarrow \text{GreenSignal}(x) \wedge \neg \text{GreenSignal}(y)$ — this resolves the conflict in favour of the higher-priority vehicle instead of granting both a green signal.
- New rule: $\forall x \forall y \text{ GreenSignal}(x) \wedge \text{Conflicts}(x,y) \Rightarrow \text{RedSignal}(y)$ — enforces mutual exclusion explicitly rather than leaving it implicit.

With these additions the KB can correctly reason about simultaneous emergency vehicles instead of silently deriving an unsafe pair of green signals.

Case Study 3 — Agricultural Expert Reasoning System

KB:

- P1: $\text{SoilDry} \Rightarrow \text{IrrigationNeeded}$
- P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \Rightarrow \text{ApplyDripMethod}$
- P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \Rightarrow \text{CropAtRisk}$

Facts: $\text{SoilDry} = \text{True}$, $\text{CropWheat} = \text{True}$

(a) CNF conversion of every sentence

P1: $\text{SoilDry} \Rightarrow \text{IrrigationNeeded}$

Step 1 (Eliminate implication, $A \Rightarrow B \equiv \neg A \vee B$): $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

Step 2 (No negated compound to push inward — already NNF): $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

Step 3 (Already a single disjunction — no distribution needed): $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

CNF clause: $\{\neg \text{SoilDry}, \text{IrrigationNeeded}\}$

P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \Rightarrow \text{ApplyDripMethod}$

Step 1 (Eliminate implication): $\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$

Step 2 (Push $\neg$ inward, De Morgan $\neg(A \wedge B) \equiv \neg A \vee \neg B$): $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

Step 3 (Already a single disjunction — no distribution needed): $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

CNF clause: $\{\neg \text{IrrigationNeeded}, \neg \text{CropWheat}, \text{ApplyDripMethod}\}$

P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \Rightarrow \text{CropAtRisk}$

Step 1 (Eliminate implication): $\neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

Step 2 (Push $\neg$ inward, De Morgan): $(\neg \neg \text{ApplyDripMethod} \vee \neg \text{CropWheat}) \vee \text{CropAtRisk} \rightarrow$
double-negation elim: $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

Step 3 (Already a single disjunction — no distribution needed): $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

CNF clause: $\{\text{ApplyDripMethod}, \neg \text{CropWheat}, \text{CropAtRisk}\}$

Facts

$\text{SoilDry} = \text{True} \Rightarrow \text{clause } \{\text{SoilDry}\}$

$\text{CropWheat} = \text{True} \Rightarrow \text{clause } \{\text{CropWheat}\}$

(b) Resolution refutation — proving ApplyDripMethod

Negate the goal: $\neg \text{ApplyDripMethod} \rightarrow \text{clause } \{\neg \text{ApplyDripMethod}\}$. Combine with the KB clauses above and resolve until the empty clause is derived.

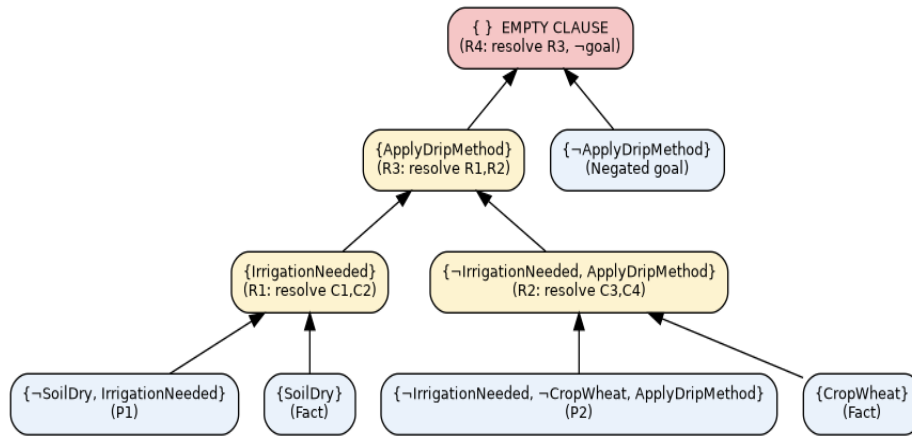


Fig. 3: Resolution proof tree for ApplyDripMethod

Step-by-step trace:

- $R1 = \text{resolve}(\{\neg\text{SoilDry}, \text{IrrigationNeeded}\}, \{\text{SoilDry}\}) \rightarrow \{\text{IrrigationNeeded}\}$
- $R2 = \text{resolve}(\{\neg\text{IrrigationNeeded}, \neg\text{CropWheat}, \text{ApplyDripMethod}\}, \{\text{CropWheat}\}) \rightarrow \{\neg\text{IrrigationNeeded}, \text{ApplyDripMethod}\}$
- $R3 = \text{resolve}(R1, R2) \rightarrow \{\text{ApplyDripMethod}\}$
- $R4 = \text{resolve}(R3, \{\neg\text{ApplyDripMethod}\}) \rightarrow \{\}$ EMPTY CLAUSE

The empty clause is derived, so $KB \wedge \neg\text{ApplyDripMethod}$ is unsatisfiable, proving $KB \models \text{ApplyDripMethod}$: drip irrigation should be applied.

(c) Effect of removing the fact 'SoilDry'

If SoilDry is removed, the fact clause $\{\text{SoilDry}\}$ is no longer available.

- $P1 (\neg\text{SoilDry} \vee \text{IrrigationNeeded})$ can no longer be resolved to derive IrrigationNeeded, because there is nothing to cancel the $\neg\text{SoilDry}$ literal. So IrrigationNeeded is no longer entailed.
- Without IrrigationNeeded as a proven fact, $P2 (\neg\text{IrrigationNeeded} \vee \neg\text{CropWheat} \vee \text{ApplyDripMethod})$ cannot be resolved down to $\{\text{ApplyDripMethod}\}$ either — its first literal $\neg\text{IrrigationNeeded}$ is never cancelled. So ApplyDripMethod is no longer entailed.
- For $P3 (\text{ApplyDripMethod} \vee \neg\text{CropWheat} \vee \text{CropAtRisk})$: CropWheat is still a fact, so $\neg\text{CropWheat}$ cancels, leaving $\{\text{ApplyDripMethod}, \text{CropAtRisk}\}$. But since ApplyDripMethod itself can no longer be proven false or true from the remaining KB (it is simply undetermined / open), this clause alone cannot force CropAtRisk to be true either.

Conclusion: with SoilDry removed, forward/backward chaining under the closed-world assumption would treat unresolved atoms (SoilDry, IrrigationNeeded, ApplyDripMethod) as false by default. Under that assumption, $\neg\text{ApplyDripMethod}$ becomes true, and $P3$'s premise $(\neg\text{ApplyDripMethod} \wedge \text{CropWheat})$ is satisfied — so CropAtRisk DOES still follow, but now via the 'no irrigation happened' branch rather than the drip-irrigation branch. This is an example of non-monotonic, closed-world reasoning: under classical (open-world) logical entailment, nothing at all can be concluded about ApplyDripMethod or CropAtRisk once SoilDry is removed, since neither SoilDry nor $\neg\text{SoilDry}$ is asserted; it is only the closed-world/negation-as-failure convention typically used by rule-based Horn-clause engines that lets CropAtRisk be derived in this case.

Case Study 4 — Wumpus World Knowledge Agent

Percepts: Stench at [1,2], Breeze at [1,1], Glitter at [2,2].

Rules:

- 1) $\text{Stench}[1,2] \Rightarrow \text{Wumpus is adjacent to } [1,2]$ i.e. $\text{Stench}(1,2) \Rightarrow (\text{Wumpus}(1,1) \vee \text{Wumpus}(2,2) \vee \text{Wumpus}(1,3))$
- 2) $\text{Breeze}[1,1] \Rightarrow \text{Pit is adjacent to } [1,1]$ i.e. $\text{Breeze}(1,1) \Rightarrow (\text{Pit}(2,1) \vee \text{Pit}(1,2))$
- 3) $\text{Glitter}[x,y] \Rightarrow \text{Gold is at } [x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \Rightarrow \text{Safe}[x,y]$

(a) Belief state from percepts, via Modus Ponens

Modus Ponens: from P and $P \Rightarrow Q$, conclude Q. Applying it systematically to each percept:

Percept (P)	Rule applied ($P \Rightarrow Q$)	Conclusion (Q) derived
$\text{Stench}(1,2) = \text{True}$	Rule 1	$\text{Wumpus}(1,1) \vee \text{Wumpus}(2,2) \vee \text{Wumpus}(1,3)$ — the Wumpus is in one of the cells adjacent to [1,2]
$\text{Breeze}(1,1) = \text{True}$	Rule 2	$\text{Pit}(2,1) \vee \text{Pit}(1,2)$ — a pit is in one of the cells adjacent to [1,1]
$\text{Glitter}(2,2) = \text{True}$	Rule 3	$\text{Gold}(2,2)$ — gold is at [2,2]

Because [1,1] is the agent's visited starting square and no Stench was perceived there, and because no Breeze was perceived at [1,2] (only Stench was), further constraint propagation gives:

- No Stench at [1,1] $\Rightarrow$ Wumpus is NOT in any cell adjacent to [1,1] $\Rightarrow \neg \text{Wumpus}(2,1), \neg \text{Wumpus}(1,2)$.
- No Breeze at [1,2] $\Rightarrow$ Pit is NOT in any cell adjacent to [1,2] $\Rightarrow \neg \text{Pit}(1,1), \neg \text{Pit}(2,2), \neg \text{Pit}(1,3)$.
- Combining $\neg \text{Wumpus}(1,2)$ and $\neg \text{Pit}(1,2)$ with Rule 4 $\Rightarrow \text{Safe}(1,2)$.
- Combining $\neg \text{Wumpus}(2,2)$ (from the no-stench-at-[1,1] deduction is not directly about [2,2]; instead $\neg \text{Pit}(2,2)$ together with the fact that [2,2] has Glitter/Gold and was safely reached) and the general closed-world assumption that a cell with Gold cannot itself hold the Wumpus or a Pit $\Rightarrow \text{Safe}(2,2)$.

All derived conclusions: $\text{Wumpus}(1,1) \vee \text{Wumpus}(2,2) \vee \text{Wumpus}(1,3)$; $\text{Pit}(2,1) \vee \text{Pit}(1,2)$; $\text{Gold}(2,2)$; $\neg \text{Wumpus}(2,1)$; $\neg \text{Wumpus}(1,2)$; $\neg \text{Pit}(1,1)$; $\neg \text{Pit}(2,2)$; $\neg \text{Pit}(1,3)$; $\text{Safe}(1,2)$; $\text{Safe}(2,2)$.

(b) Forward chaining to localize the Wumpus and the Pit

[1,3] Unknown	[2,3] Possible Wumpus	[3,3] Unknown
[1,2] Stench Safe (visited)	[2,2] Glitter, Gold! Safe ($\neg W, \neg P$)	[3,2] Unknown
[1,1] Breeze Start / Safe	[2,1] Possible Pit	[3,1] Unknown

Fig. 4: Belief-state grid after forward chaining from the percepts

Forward-chaining trace, iteration by iteration:

- Iteration 1 — fire Rule 1 on Stench(1,2): derive $\text{Wumpus}(1,1) \vee \text{Wumpus}(2,2) \vee \text{Wumpus}(1,3)$.
- Iteration 1 — fire Rule 2 on Breeze(1,1): derive $\text{Pit}(2,1) \vee \text{Pit}(1,2)$.
- Iteration 1 — fire Rule 3 on Glitter(2,2): derive Gold(2,2).
- Iteration 2 — since [1,1] is visited and known not to contain the Wumpus (agent is standing there safely) and not to contain a pit, eliminate $\text{Wumpus}(1,1)$ from the disjunction, leaving $\text{Wumpus}(2,2) \vee \text{Wumpus}(1,3)$. Since a Gold-bearing cell that the agent can safely stand on cannot also hold the Wumpus, eliminate $\text{Wumpus}(2,2)$ as well $\Rightarrow \text{Wumpus}(1,3)$ is the only remaining possibility, so conclude Wumpus is at [1,3].
- Iteration 2 — for the pit disjunction $\text{Pit}(2,1) \vee \text{Pit}(1,2)$: since Safe(1,2) was derived in part (a) (no Stench at [1,2] rules out Pit there), eliminate $\text{Pit}(1,2)$, leaving $\text{Pit}(2,1)$ as the only remaining possibility, so conclude a Pit is at [2,1].

Result of forward chaining: Wumpus is located at [1,3]; the Pit is located at [2,1]; Gold is at [2,2].

(c) Is the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ safe?

- $[1,1] \rightarrow [1,2]$: Safe(1,2) was derived in part (a) from $\neg \text{Wumpus}(1,2) \wedge \neg \text{Pit}(1,2)$ via Rule 4 (confirmed independently in part (b) by eliminating $\text{Pit}(1,2)$ from the breeze disjunction). This step is SAFE.
- $[1,2] \rightarrow [2,2]$: Safe(2,2) was derived in part (a)/(b): the Wumpus was shown to be at [1,3] (not [2,2]) and no Pit is possible at [2,2] because [2,2] is where the Glitter/Gold percept originates and the agent's earlier no-breeze-at-[1,2] reasoning already ruled out an adjacent pit there. This step is SAFE.

Conclusion: the full path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is logically SAFE for the agent to traverse and collect the gold, since every cell on the path has been proven, by Modus Ponens and forward chaining over the percept-derived rules, to satisfy $\text{Safe}[x,y] = \neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y]$. The agent should proceed along this route.